Design Rationale â VARE Breast-Cancer Risk Avatar

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Problem and goal

Breast-cancer risk calculators return a percentage many users find meaningless or frighteningâ neither drives reliable follow-up. VARE adds a 2-3 minute conversational layer after a Gail-inspired educational estimate: explain the number, adapt to understanding/emotion/barriers/readiness, support decision preparedness, and motivate optional next stepsâ without diagnosing or impersonating a clinician. Educational prototype only; not clinically validated.

Design decision: adaptive orchestration

The pipeline separates conversational-state estimation, Ottawa-inspired decisional-needs support, deterministic theory-based strategy selection, vetted local RAG, dynamic Groq wording (with local fallback), and safety validation. Groq never chooses strategy or invents medical facts outside retrieved evidence.

Theories and operationalization

Fuzzy-Trace Theory (FTT) â Reyna 2008; Wolfe et al. 2015; Widmer et al. 2015. Users need gist (probability is not diagnosis) before action talk. Implemented via clarify_risk + teach-back when understanding is partial/incorrect.

Health Belief Model (HBM) â Rosenstock 1974. Preventive action requires calibrated severity, benefits vs barriers, self-efficacy, cues to action. Elevated branch calibrates severity and surfaces barriers; average branch reassures without false certainty.

Motivational Interviewing (MI) â Rollnick et al. 2008; Mercado et al. 2023. Direct persuasion increases resistance; reflection and open questions preserve autonomy in acknowledge_emotion and planning turns.

Ottawa Decision Support Frameworkâ inspired â Stacey et al. 2014. Estimates temporary decisional needs (missing information, unclear options/preferences, low confidence) and supports drafting/timing without making medical decisions. Selected options (portal vs phone) persist across turns.

Risk-communication framing â Fagerlin et al. 2007; NCI 2024. Natural frequencies (â X in 100â) over relative-only framing; grounded in vetted NCI evidence chunks.

Avatar choice

Production: LiveAvatar â Ann Doctor Sittingâ â professional, approachable female presenter; warm tone aligned with NCI/ACS patient-education style (ages 35-85). Failover: static â Mayaâ image + text when video unavailable. Role: AI health educator, not a clinician (reduces authority misattribution). Architecture: LiveAvatar LITEâ app controls all medical content; avatar lip-syncs verbatim validated text only. Wolfe 2015 supports FTT-grounded avatar risk tutoring; Mercado 2023 supports MI-consistent embodied agents.

Engineering and safety

LiveAvatar LITE (1 credit/min) preserves RAG/theory control vs FULL. Keyword RAG: 17 vetted chunks (NCI, USPSTF, ACS). Gail-inspired form (not official BCRAT). Cloudflare Workers single deploy. Consent + fixed safety overrides for diagnosis, treatment, urgent symptoms, crisis.

References

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